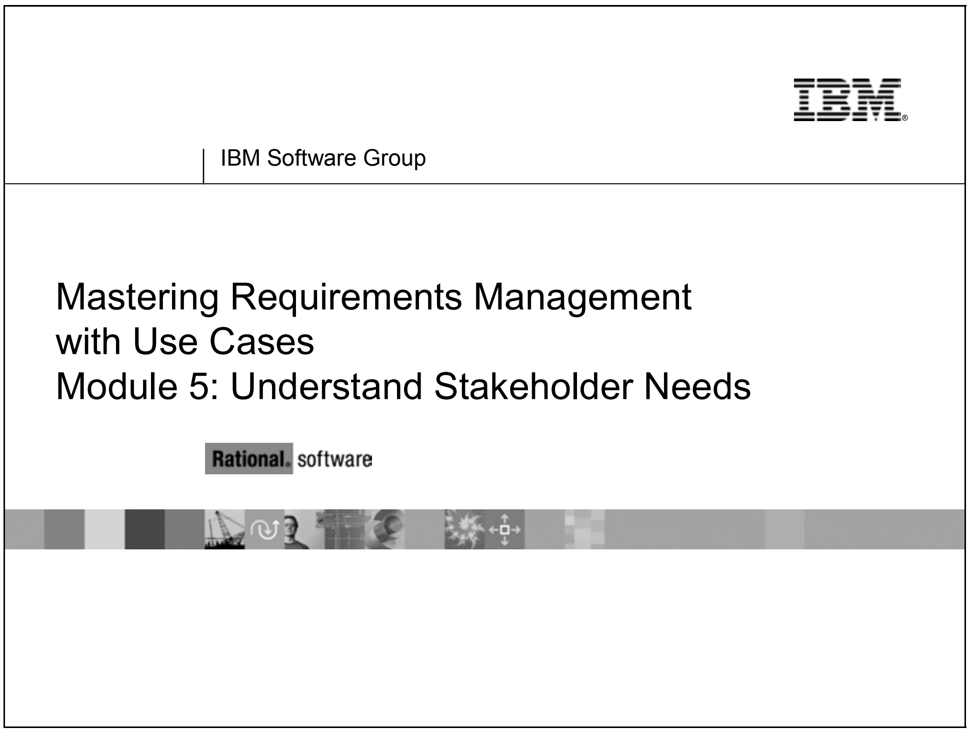
Module 5
Understand Stakeholder Needs



Topics

Objectives: Understand Stakeholder Needs

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- Identify sources for stakeholder requests.
- Describe the Stakeholder Request Artifact.
- List techniques to elicit stakeholder requests.
  - Practice brainstorming techniques.
  - Identify requirements from a customer-generated specification.

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At this point, you work on making sure that you have considered all stakeholder requests, what needs are addressed by these (relating to the problem we agreed to address), and what features in the system we might use to address these needs.

Stakeholder requests come in at many different levels of detail, often not expressed as a real need, but sometimes as a feature of the system or even as a change to a software requirement.

In this module, you discuss how to effectively communicate with your stakeholders to capture their real needs.

Where Are We in the Requirements Discipline?

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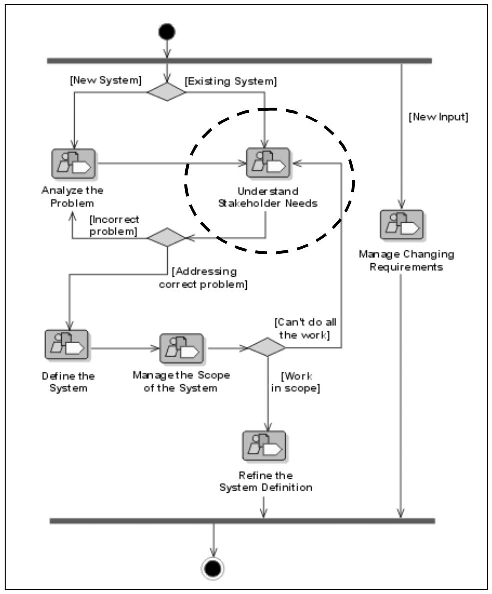

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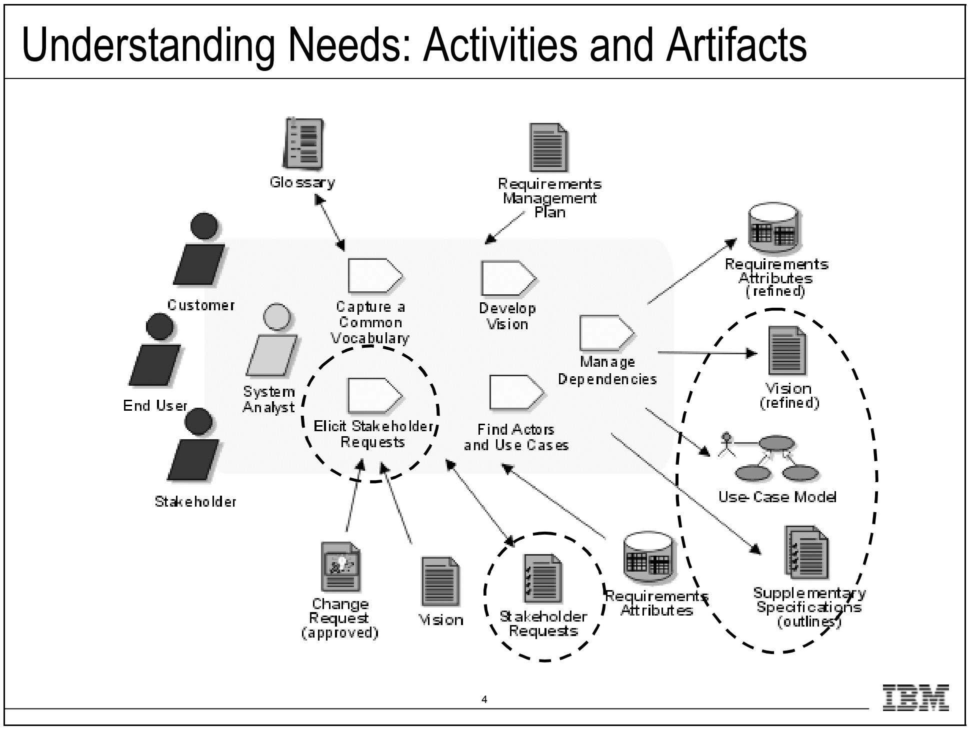
Where does the Understand Stakeholder Needs detail workflow fit in our development process? The Workflow detail “Understanding Stakeholder Needs” in the RUP® represents activities that you undertake early in the requirements gathering process.

Where is this workflow detail done in your own software development process?

Understanding Needs: Activities and Artifacts



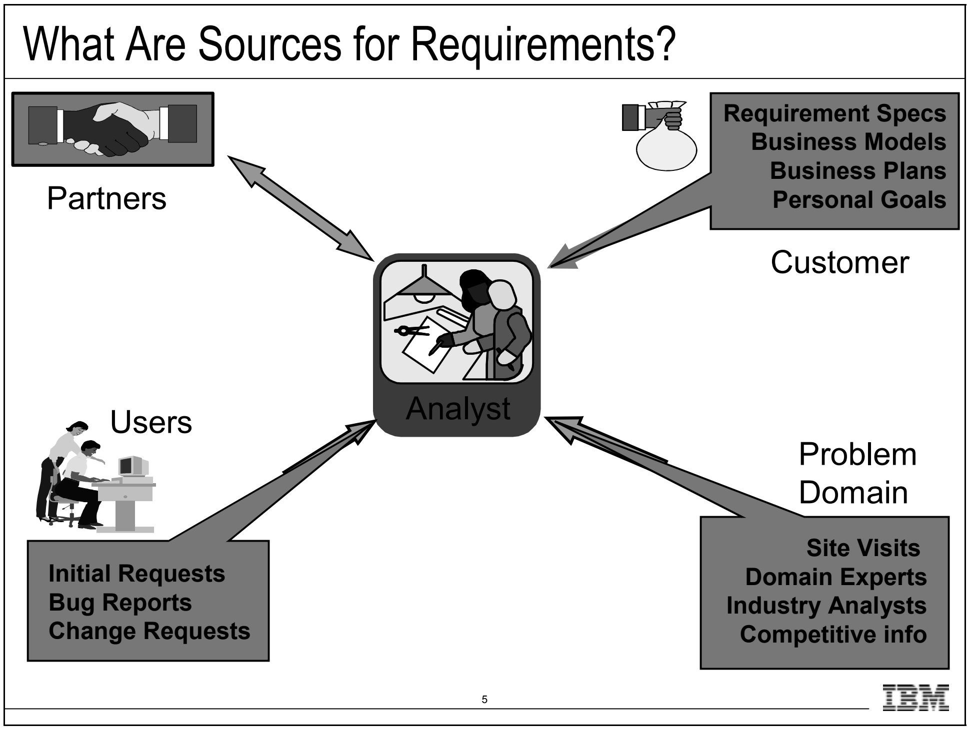
In RUP®, the objective now is to collect and elicit information from stakeholders in the project. This information can be regarded as a "wish list" that is used as primary input in defining use cases and supplementary requirements.

This workflow detail is performed mainly during iterations in the Inception and Elaboration phases. However, stakeholder requests should be gathered throughout the project by reviewing change requests following your Change Request Management process. You should be aware of incoming stakeholder requests throughout the product lifecycle, to make sure you continue to address your stakeholders' real needs.

The key activity is to elicit stakeholder requests and determine the requested features of the system. The primary outputs are a collection of prioritized Stakeholder Needs and Features, as well as the attributes associated with them. These outputs are recorded in the Vision document. Also, during this workflow, you discuss the system in terms of its use cases and actors.

Another important output is an updated Glossary of Terms to facilitate a common vocabulary among team members.

What Are Sources for Requirements?

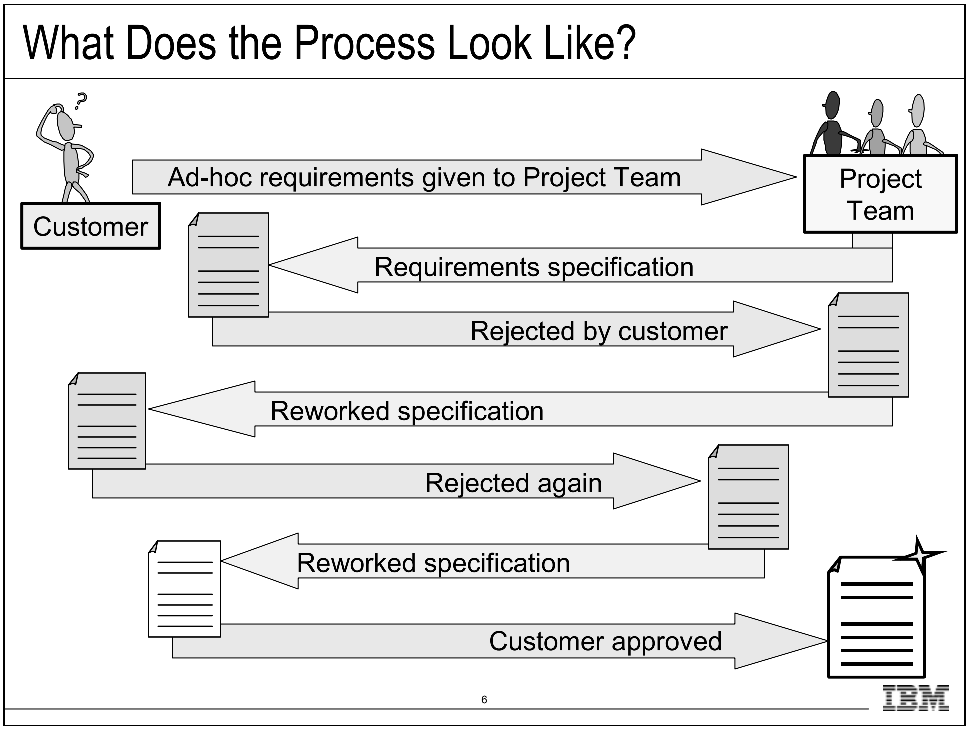


You need to capture requests from all stakeholders and specify how these requests are addressed. Although the system analyst is responsible for gathering this information, many people contribute to it: marketing people, users, customers- anyone who is considered to be a stakeholder in the project.

Other examples of external sources for requirements are:

- Statement of work
- Request for proposal
- Mission statement
- Problem statement
- Business rules
- Laws and regulations
- Legacy systems
- Business models
- Any results from requirements elicitation sessions and workshops

What Does the Process Look Like?



The traditional approach to developing requirements specifications is to strive to get it right the first time and almost always failing in some fundamental way. Modern software development practices recognize that the specifications will almost constantly be wrong in some way, and identify ways to best mitigate the risk as the project goes forward.

Continuous discussions should occur between development and the customer until an agreement is reached on the requirements of the system.

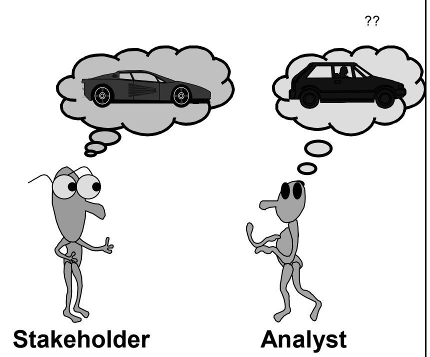
Expect to rework the requirements until you gain agreement with the customer and the other stakeholders in the project.

Some people tend to see each rejection as a failure to get the requirements correct. We encourage you to think instead of the advantages of having the requirements rejected: every time your specifications are rejected, you get new information. When a client says what they DO NOT LIKE, they are giving you important information about what they want instead. This process of rejection helps the client focus on what is important to their organization.

What Problems Might Be Encountered?

What Problems Might Be Encountered?

- Stakeholders
  - Have a pre-conceived idea of the solution.
  - Do not know what they really want.
  - Are unable to articulate what they want.
  - Think they know what they want, but do not recognize it when it is delivered.
- Analysts
  - Think they understand user problems better than users.
- Everybody
  - Everyone sees things from their own point of view.
  - Believes everyone is politically motivated.



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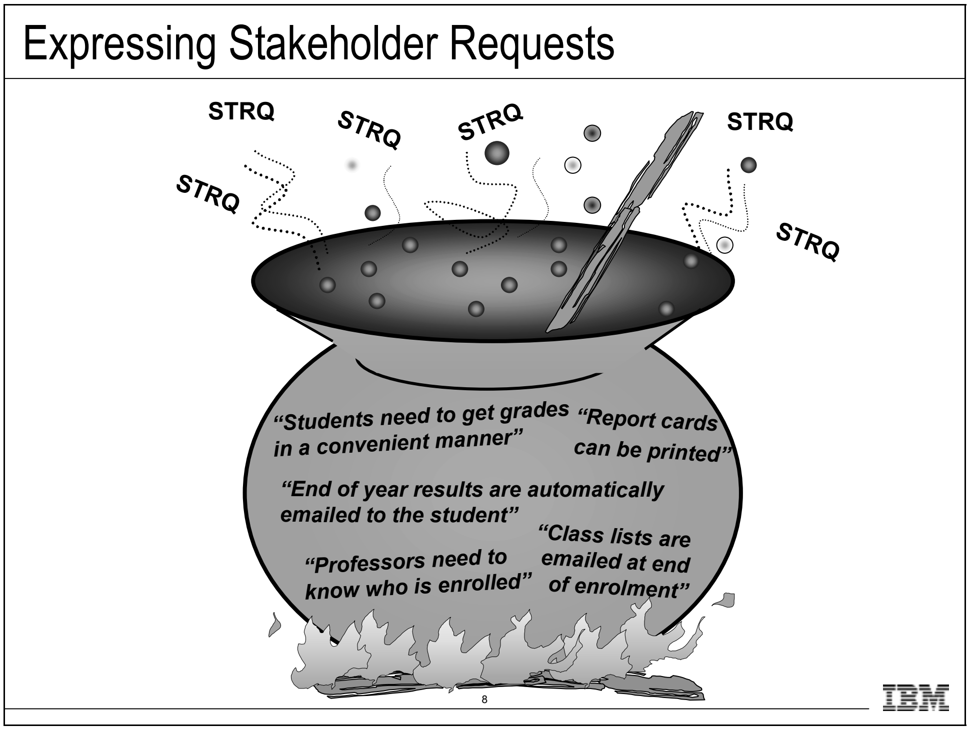
What are some of the problems you have encountered in determining your stakeholder needs?

Expressing Stakeholder Requests



Jumping to solution mode is a common problem we humans have. It is because of this that, when you elicit needs from your stakeholders, you often get them expressed as features instead of a statement of what the stakeholder needs from any solution in order to solve the business problem. What do you do with these needs expressed as features? Do you tell the customer to "hold off – we'll get to those later"? No. At this stage, it is a stakeholder request and should be recorded as one. At the appropriate time you should assess whether it is valid and then create a feature requirement for it. The text of the new feature may be identical to the text of the stakeholder request. (In the USA DoD standards, the process of copying the text of a requirement to the next level is called *allocation*. If you reword the requirement when you create it at the next level, it is called *derivation*.)

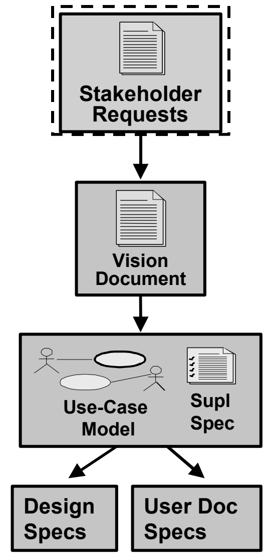
To understand the need behind a request expressed as a feature, you can perform some problem analysis. For example: "The defect tracking system shall provide a project status trending report." This is a stakeholder request expressed as a feature. To understand the need that drives this feature, ask "why?" The answer could be something like: "I need to be able to understand if defects being raised faster than they are being resolved." This is the real need behind the feature, a sub-problem if you like.

Either way, you should record every request and then send it through a standard approval process before it is included in the system. Stakeholder requests expressed as features require a little more analysis to test whether they are valid and that they support the business goals of the system.

The Stakeholder Requests Artifact

The Stakeholder Requests Artifact

- Is owned by the stakeholders.
- Contains all requests from the stakeholders.
- Is consolidated from many sources.
  - E-mail, customer requirements specification, napkins, white boards, spreadsheets, and so on.
- Used by project team to derive features and software requirements.
- May contain references to any type of external source.



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Stakeholder requests come from many sources in many different forms. The most common form is a customer requirements specification. But requests also come via email, meetings, informal discussions over lunch, etc. All of these requests need to be recorded somewhere and negotiated for inclusion into the project.

One way is for your stakeholder or Product Champion to add it to the requirements management repository as a stakeholder request. It would immediately have a status of “proposed.” The stakeholder requests must then go through a formal negotiation for inclusion in the project. (Failure to do this could lead to massive scope creep.)

One problem you will have is that some requests come in documents (usually earlier in the project), and then other requests come in different forms, such as email. Managing these different forms can be difficult. A solution to this is to maintain all stakeholder requests in a database. If you are using a tool like RequisitePro, adding the requirements to the database from a document is a simple process. Subsequent work directly in the database is also a simple task.

Techniques for Eliciting Stakeholder Requests

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- Review customer requirement specifications
- Requirements workshop
  - Use-case workshops
  - Brainstorming and idea reduction
- Interviews
- Questionnaires
- Role playing
- Prototypes
- Storyboards

Appendix

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These elicitation techniques are useful for gathering information about stakeholder needs. The techniques can also be used very effectively for gathering information about feature requirements or detailed software requirements.

Many of these techniques can be used together. For example, a requirements workshop brings stakeholders together. At the requirements workshop, the participants may brainstorm for new ideas, or they may sketch out the use cases for the system to be built.

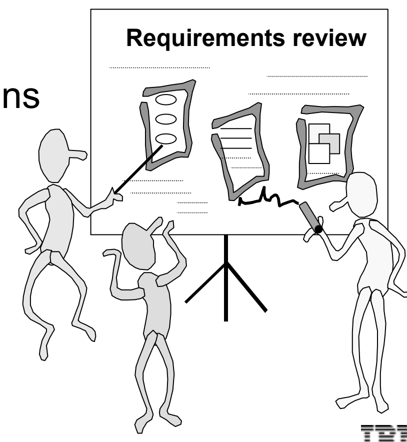
There are many excellent resources available to learn more about each of these elicitation techniques. The following list provides just a few resources that you can use:

- Requirements Workshop: *Requirements By Collaboration*. By Ellen Gottesdiener. Addison-Wesley. ISBN: 0-201-78606-0.
- Interviews: *Exploring Requirements: Quality Before Design*. D. Gauss & G. Weinburg. Dorset House. ISBN: 0-932-63313-7.
- Questionnaires: *Exploring Requirements: Quality Before Design*. D. Gauss & G. Weinburg. Dorset House. ISBN: 0-932-63313-7.
- Role Playing: *Child's Play: Using Techniques Developed to Elicit Requirements from Children with Adults.* N. Millard, P. Lynch, and K. Tracey., Proc. of 3rd Int'l. Conf. on Requirements Eng. (ICRE '98), Colorado Springs, USA, April 6-10, 1998.

Review Customer Requirement Specifications

Review Customer Requirement Specifications

- Identify requirements.
  - Recognize and label:
    - Application behaviors
    - Behavioral attributes
    - Issues and assumptions
- Ask the customer.



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Project teams sometimes receive their requirements in a document directly from their customer, especially when contracting to do a job.

In this case, it is important to review the customer's specification to determine what you believe to be the stated requirements. The identified requirements then need to be taken back to the customer to be verified; any issues noted must be addressed and resolved.

Keep in mind who wrote the specification. The specification is inevitably written from the author's perspective.

Introductory and overview sections of the specification are important for general understanding of the specifications. Those sections are less likely to contain requirements, but keep in mind there might be misplaced requirements.

Exercise 5.1: Review Customer Requirements Spec

Exercise 5.1: Review Customer Requirements Spec

- Part 1
  - Review the customer requirements specification.
    - RU e-st Requirements Specification.
  - Look for possible requirements in the spec.
- Part 2
  - Review the list of sample stakeholder requests.
  - Refine the Vision document.
    - Define the system boundary.
    - Revise the list of actors.

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See Student Workbook Exercise 5.1: Review Customer Requirements Specification.

Brainstorming

Brainstorming

- Generates as many ideas as possible.

Rules for Brainstorming

- Clearly state the objective of the session.
- Generate as many ideas as possible.
- Let your imagination soar.
- Do not allow criticism or debate.
- Combine ideas.

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Brainstorming is a way to quickly generate as many ideas as possible. In the idea generation phase, the emphasis is on quick generation of ideas. In the idea reduction phase, the emphasis is on combining ideas and focusing on the most promising ideas. In order to encourage people to contribute ideas, it is important to finish with idea generation before doing any pruning or criticizing of ideas. Ideas may start to change as people hear ideas and come up with variations on them. This is good because some of the best ideas are those that build on the expertise and creativity of many people in the group.

One technique: This is a four-step process. None of the steps should be skipped!

- **Brainstorm** the Features that the new Software System should realize. Cover a wall with yellow sticky notes.
- **Classify the Stickies:** *This process must be done with no talking!* Invite everyone to move around the stickies and group those that describe a single feature. Additional stickies can be added at this time, but no one can talk to anyone except to the Facilitator.
- **Name that Feature:** the facilitator then looks at each group of stickies, reads the stickies out loud, and helps the group name that feature. Affinitizing only gets you 60-80 percent in the right groups. Naming the groups realigns the groups. Get rid of some and split up others. You almost always end up with 12-17 groups.
- **Prioritize the Features:** It is critical to understand which are the key features that the software must support. Use a process, such as bullet voting. It is quick and easy and almost always ends up with a clear bell curve.

Brainstorming Advantages and Disadvantages

Brainstorming Advantages and Disadvantages

- Advantages
  - Used anytime, anywhere.
  - Good for groups.
  - Good for high-level entities and assumptions.
  - Amenable to some automation.
- Disadvantages
  - Susceptible to group processes.
  - Unsystematic in "classic" form.

Takeda et al. 1993

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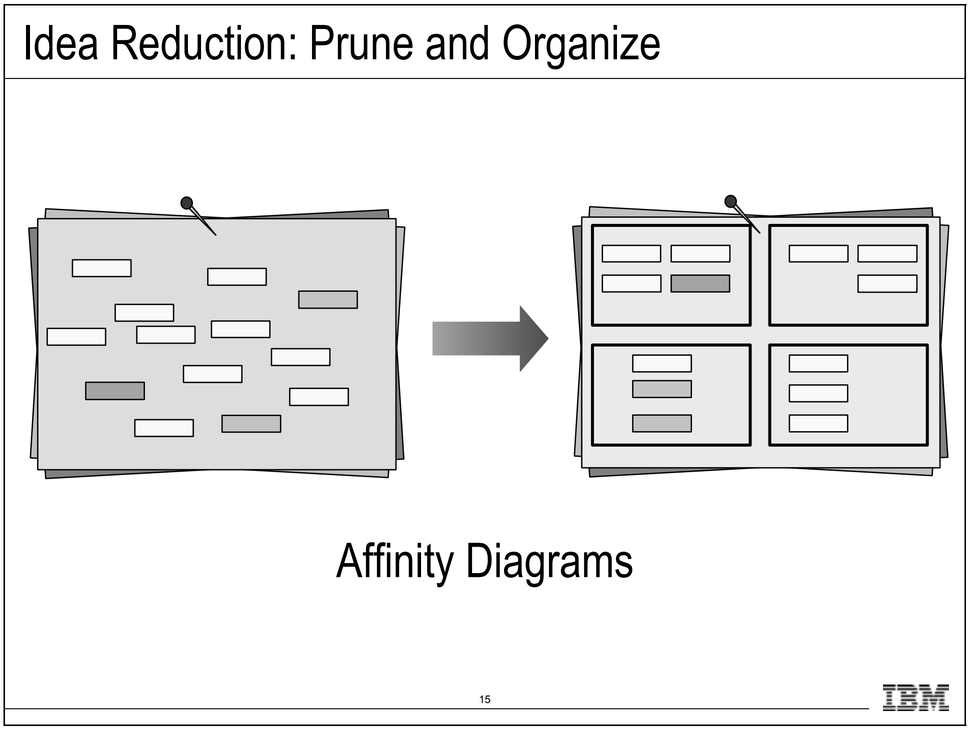
Description: the requirements engineer asks a group of stakeholders to generate as many ideas as possible with an emphasis on generation rather than on evaluation.

Preconditions: a suitable group of stakeholders.

Strengths: good for eliciting high-level domain entities and questioning assumptions which might otherwise have constrained approaches considered. Tool support available for some varieties.

Weaknesses: susceptible to group processes; unsystematic in "classic" form, though some varieties overcome this.

Idea Reduction: Prune and Organize



The affinity diagram is used to organize the results of a brainstorming session into manageable groups. It is based on the relationship between the items found during brainstorming. It is a creative, rather than logical process.

After all the ideas are collected on the board the participants get up and move the cards or notes into groups that appear to be similar ideas. It is important that this be done silently. A rough guideline is that you will find seven to ten groups. Continue this activity until everyone agrees on the grouping.

The participants then discuss the relationships between the grouped items and identify a name for each group of cards or notes. Often one of the cards within the group serves as a title for the entire group; if not, create one and place it over the group.

If there are any items that fell into a miscellaneous group, see if they now fit one of the designated groups.

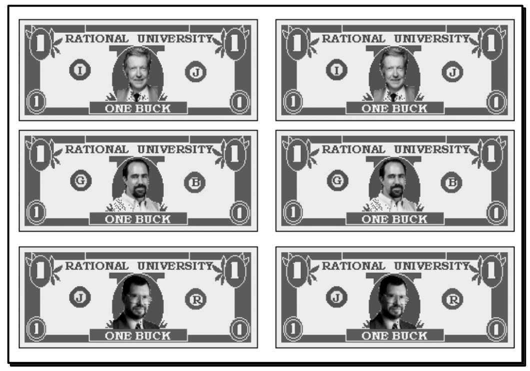
Those titles that were selected for the groups can now be studied to gain a better understanding of the original problem.

A good reference for affinity diagrams is: H. Beyer and K. Holtzblatt, "Contextual Design: Defining Customer-Centered Systems," Morgan Kaufmann Publishers, 1997.

Idea Reduction: Prioritize Ideas

Idea Reduction: Prioritize Ideas

- Prioritize remaining ideas.
  - Vote
    - Cumulative votes
      - Buy ideas
    - Single vote
  - Apply evaluation
  - criteria
    - Non-weighted
    - Weighted



Rational University "bucks"

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Now, what do you do after you have gathered all these "great ideas"?

After the idea gathering and pruning are over, organize and evaluate them.

There are many ways to evaluate them. One technique you can use to generate a pareto-like diagram is to use a process where the group "buys features". *Buying ideas* is a way to do cumulative voting. Each person in the requirements workshop would receive a fixed number of dollars. A person gives one "vote" for an idea by putting a dollar on a particular idea.

For example, use cumulative voting, where each person gets three votes. People raise their hands for each item they vote for. The facilitator writes the number of votes on each idea's post-it note.

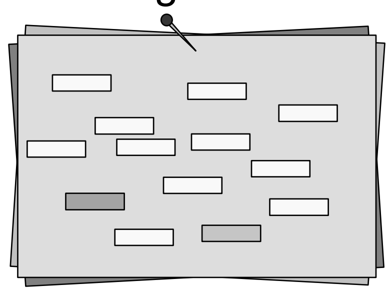
This method can be abused if someone puts all their money on their item which is unpopular with everyone else.

An alternative to voting is to apply names such as "critical", "important", "useful" and weighting the 9-3-1.

Exercise 5.2: Brainstorming

Exercise 5.2: Brainstorming

- Gather ideas for stakeholder requests/needs.
- Clarify and organize the ideas.
- Condense ideas.
- Prioritize remaining ideas.



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See Student Workbook Exercise 5.2: Utilize Brainstorming Techniques.

Assume the roles of different stakeholders you identified in Exercise 4.2 and think about what they would need from the system that would help solve their problem.

To get you started, here is a short list of of stakeholder needs:

- Trading Customer: "Obtain current and historical quotes "
- Trading Customer: "Maintain personal portfolio"
- Trading Customer: "Trade anytime from anywhere"
- Trading Customer: "Secure environment"
- Marketing: "Advertise additional services"

Considerations for Selecting Elicitation Techniques

Considerations for Selecting Elicitation Techniques

- Requirements Purpose
  - Specification for design and implementation.
  - Selecting off-the-shelf packages.
  - Legal contract for system procurement.
- Knowledge Types
  - Different methods acquire different types of knowledge.
- Internal Knowledge Filtering
  - Some knowledge can be retrieved from memory; whereas, other knowledge cannot.
- Acquisition Context
  - Environment can influence elicitation techniques.

WP6:
ACRE:
Selecting
Methods for
Requirements
Acquisition

Maiden N.A.M. & Rugg G., 1996

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The ACRE Framework argues that “more than one acquisition method is needed to capture the full range of complex requirements for most complex software-intensive systems.” The framework considers methods from software engineering, knowledge engineering, and social sciences.

The purpose of your requirements gathering efforts determines the format you want the requirements represented. Different techniques are used depending upon how you want the requirements expressed. For example, if capturing requirements for selecting commercial packages you want requirements captured in a format so they can be compared. If you are capturing for the purposes of a legal contract, you will want requirements expressed in a natural language format as this is how these documents are written.

Knowledge can be categorized into behaviour, process, and data. Acquiring data knowledge is difficult because stakeholders are typically not aware of the information around them. Special methods should be used to acquire data knowledge.

When capturing requirements about existing domains, people unconsciously filter their knowledge. Some knowledge can be easily retrieved from memory (non-tacit), and other knowledge cannot (tacit). For example, tacit knowledge is knowledge that has become habitualized. You must select your elicitation techniques carefully to obtain all the knowledge.

Acquisition context touches the social aspect of requirements acquisition. Thought needs to be given to things, such as power positions in the organization, resource availability, technological support. Elicitation techniques should be tailored with these constraints in mind.

Brainstorming ACRE Classification Example

Brainstorming ACRE Classification Example

Purpose	Known COTS	-
	Unknown COTS	-
	System Procurement	✓
	System Development	✓✓
Knowledge Type	Behavior	✓
	Process	✓
	Data	-
Effectiveness of brainstorming for acquiring knowledge with different internal representations	Future System	✓
	Non-Tacit	✓
	Recognized	✓
	Taken-for-granted	-
	Working memory	X
	Compiled	✓✓
	Implicit	-
	Observable	X
Conditions for Method Use	Meeting Needed	✓
	Time to Prepare	✓✓
	Time for Session	✓✓
	Time to Obtain	✓
	Number of Stakeholders	✓
	Friendliness	✓
	No Technologies	✓✓

LEGEND

- ✓ **Good fit**
- ✓✓ **Very good fit**
- X **Weak fit**
- \- **Poor fit**

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The table is an example of how brainstorming fits into the ACRE Framework. For further details, refer to WP6 in the Student Workbook.

Note: COTS – Common Off-The shelf Software

Review: Understand Stakeholder Needs

Review: Understand Stakeholder Needs

1. What are some elicitation techniques for understanding user needs?
2. What is the relationship between a need and a feature when expressed by a stakeholder during Understand Stakeholder Needs?
3. What should you do with a need that is expressed as a feature?
4. What does the ACRE Framework say about requirements elicitation?

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